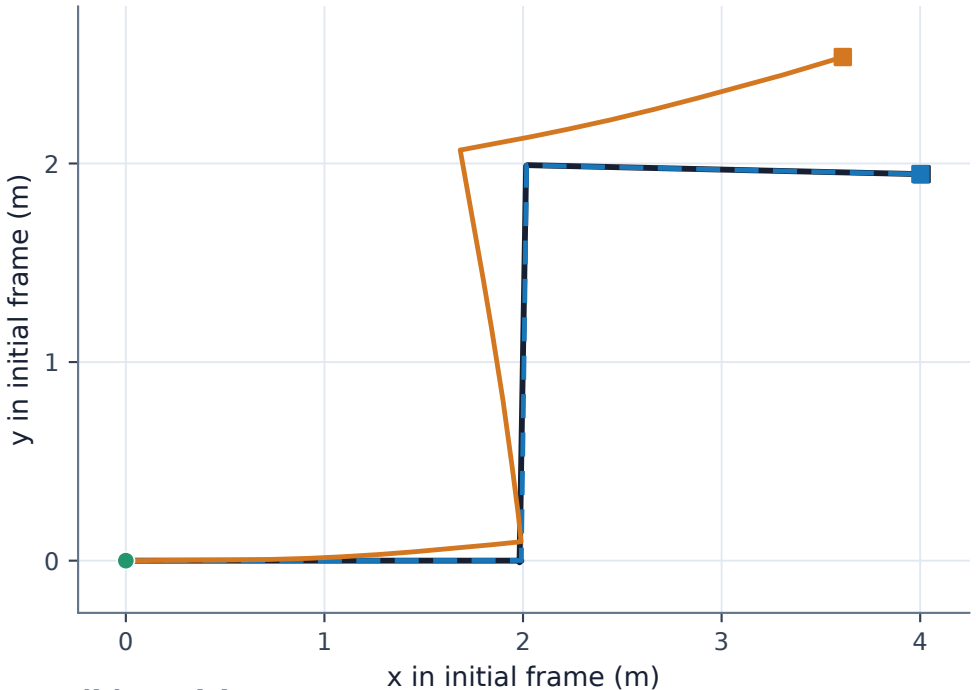
Online Wheel Measurement Uncertainty

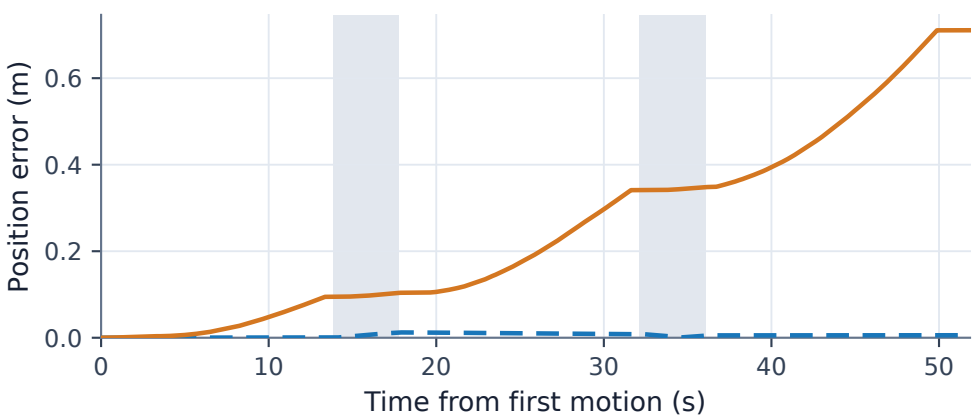
Baseline and combined-error odometry | Independent simulator ground truth

(a) Planar trajectories

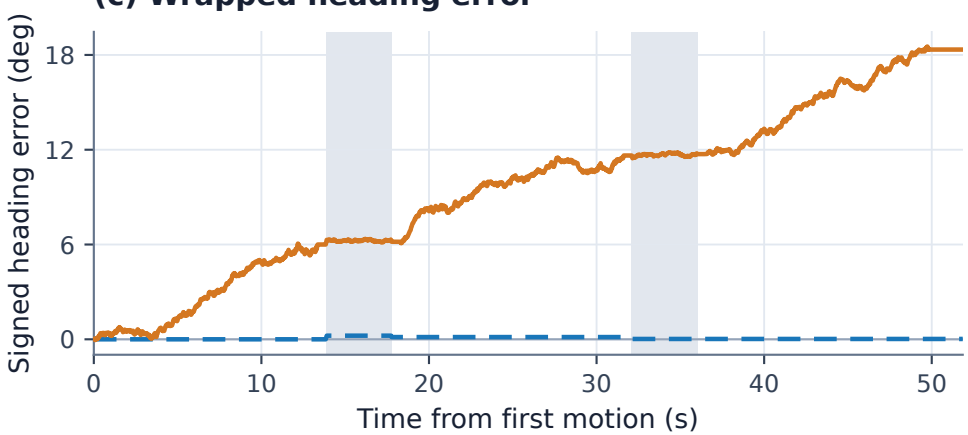


	Baseline	Online A3
Position RMSE (mm)	**6.84**	**327.23**
Endpoint error (mm)	**5.77**	**710.78**
Heading RMSE (deg)	**0.098**	**10.768**

(b) Position error



(c) Wrapped heading error



One recorded simulation run; paired baseline and A3 (n = 1 each). Shaded intervals: commanded turns.

A3 measurement assumptions: sigma_1m = 0.002 m; left bias = -0.5%; right bias = +0.5%; seed = 0. Coefficients are illustrative.

One fixed initial SE(2) alignment per path; ground truth is used only for evaluation. Heading errors are wrapped to [-180 deg, 180 deg].

A fixed seed reproduces only the same accepted input sequence; this online path need not match an earlier offline replay.